SYSTEM AND METHOD FOR ENCRYPTED KEYBOARD EVENTS

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Title: SYSTEM AND METHOD FOR ENCRYPTED KEYBOARD EVENTS

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Abstract

A method and system for enhancing the security of text input by encrypting each individual keystroke event independently. The system comprises a keyboard (physical or virtual) that captures individual keystroke events and a cryptographic processing unit that encrypts each event as a distinct ciphertext before transmission over a data bus or network. This prevents an adversary from performing cryptanalysis on contiguous blocks of text, which is a common vulnerability exploited by audio side-channel attacks on physical keyboards and timing analysis on software keyboards. This approach significantly increases the computational effort required for decryption by state-sponsored malicious actors, even when augmented with machine learning and advanced surveillance techniques.

Background of the Invention

Sophisticated adversaries, particularly state-sponsored malicious actors, frequently employ advanced cryptanalysis techniques to compromise encrypted communications. A prevalent

method involves side-channel attacks, such as acoustic analysis of keystrokes on physical keyboards or timing analysis of input events on software keyboards (e.g., in web applications like Gmail or messaging apps like WhatsApp). By analyzing the timing and sequences of keystrokes, malicious actors can infer characteristics of the typed text, such as overall length, number of words, and individual word lengths. This metadata, when combined with machine learning models trained on a user's lexicon or past writings, can exponentially increase the effectiveness of cryptanalysis, leading to the rapid compromise of encryption keys for entire communication sessions.

Summary of the Invention

The present invention provides a novel method and system for securing text input by fundamentally altering the granularity of encryption from messages or text blobs to individual keystroke events. This paradigm shift dramatically mitigates the effectiveness of side-channel cryptanalysis.

The system, applicable to both physical keyboards (e.g., connected via an encrypted bus) and virtual/software keyboards (e.g., within web applications or mobile apps), captures each distinct keystroke as an independent event. A dedicated cryptographic unit then encrypts each individual keystroke, generating a unique ciphertext for every key press. These discrete ciphertexts are transmitted separately over the communication channel (bus or network). This ensures that an adversary cannot observe patterns within contiguous blocks of encrypted text or analyze timing variations to infer meaningful metadata about words or sentences. Consequently, machine learning models relying on such metadata for cryptanalysis become largely ineffective, as each keystroke provides minimal context. While this approach may increase bandwidth usage slightly, particularly for software keyboards, the enhanced security benefits far outweigh this minor overhead in modern high-speed networks.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and

local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the system for encrypting individual keystroke events from a keyboard.

FIG. 2 is a flowchart depicting the method for processing and encrypting individual keystroke events.

FIG. 3 is a conceptual diagram illustrating the application of individual keystroke encryption in a software keyboard environment.

Detailed Description of the Invention

Referring to **FIG. 1**, the system (100) for encrypted keyboard events is shown. A Keyboard (110), which can be a physical hardware keyboard, generates individual Keystroke Events (111). These events are sent to a Cryptographic Unit (120), which encrypts each event independently, producing Individual Ciphertexts (121). These ciphertexts are then transmitted over an Encrypted Bus or Network (130) to a receiving application (140), where they are decrypted (141) and reassembled into plaintext (142).

Referring to **FIG. 2**, the method (200) for processing and encrypting individual keystroke events is illustrated. The process begins with 'Detect Individual Keystroke Event (210)'. This is followed by 'Generate Unique Ciphertext for Keystroke (220)'. The ciphertext is then 'Transmit Individual Ciphertext (230)'. These steps are repeated for each keystroke, preventing the aggregation of plaintext information for cryptanalysis.

Referring to **FIG. 3**, the application in a software keyboard environment (300) is depicted. A User (301) interacts with a 'Software Keyboard (302)' (e.g., in a web browser or mobile app). Each virtual keypress generates a 'Keystroke Event (303)'. This event is processed by a 'Client-Side Cryptographic Module (304)', which encrypts it into an 'Individual Ciphertext (305)'. The ciphertexts are then sent over an 'Encrypted Network (306)' to a 'Server (307)' for decryption and processing. This ensures that even in a web-based context, each keystroke is protected against timing and content analysis.

Claims

What is claimed is:

1. A method for securing text input, comprising:

    - detecting an individual keystroke event from an input device, wherein the input device is a physical keyboard or a software keyboard;
    - encrypting the individual keystroke event to generate a distinct ciphertext corresponding solely to that keystroke event; and
    - transmitting the distinct ciphertext over a communication channel.

2. The method of claim 1, wherein the communication channel is a data bus or a network.

3. The method of claim 1, further comprising receiving a plurality of distinct ciphertexts, each corresponding to an individual keystroke event, and decrypting and reassembling said ciphertexts into plaintext.

4. The method of claim 1, wherein the encryption of each individual keystroke event prevents cryptanalysis based on timing analysis or content analysis of contiguous blocks of text.

5. The method of claim 1, wherein the software keyboard is implemented within a web application or a mobile application.

6. A system for securing text input, comprising:

    - an input device configured to detect individual keystroke events, wherein the input device is a physical keyboard or a software keyboard;
    - a cryptographic processing unit communicatively coupled to the input device, the cryptographic processing unit configured to encrypt each individual keystroke event into a distinct ciphertext; and
    - a transmission module configured to transmit each distinct ciphertext over a communication channel.

7. The system of claim 6, further comprising a reception and decryption module configured to receive, decrypt, and reassemble the distinct ciphertexts into plaintext.

Drawings

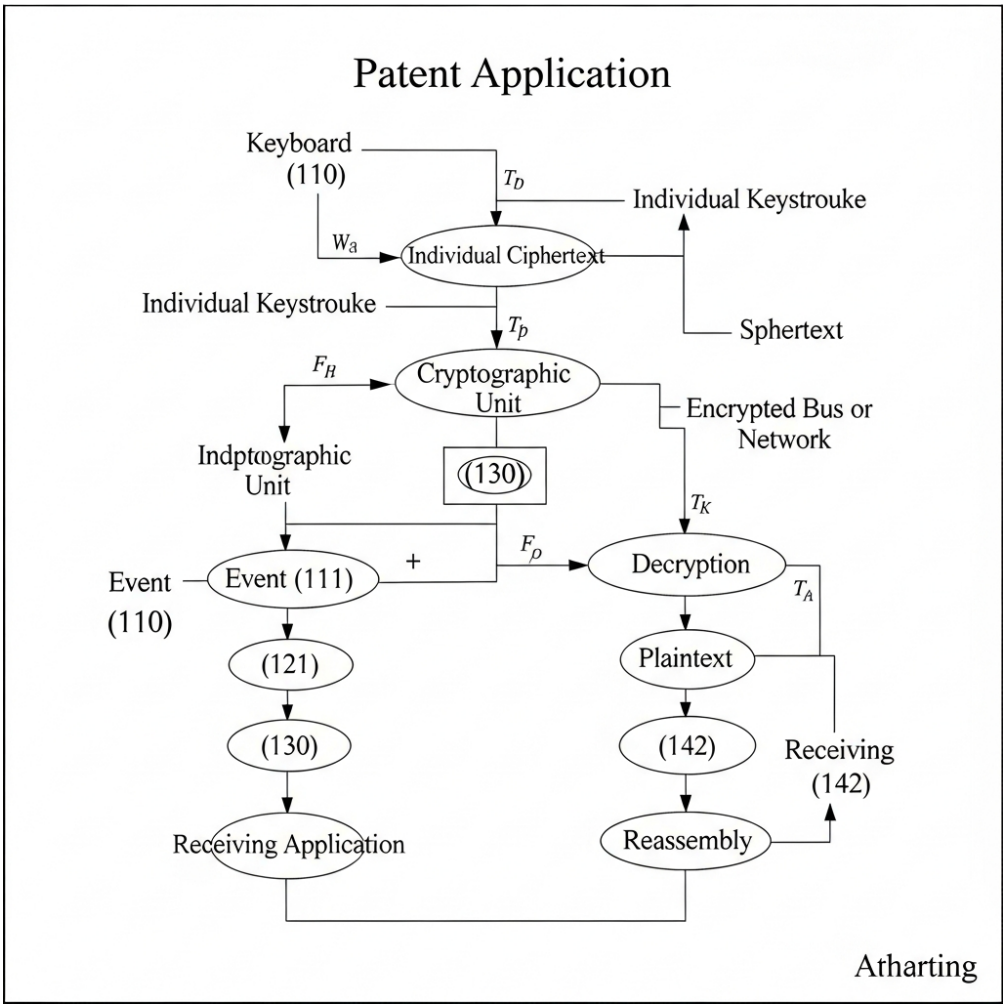


Figure 1: Block diagram illustrating the system for encrypting individual keystroke events from a keyboard.

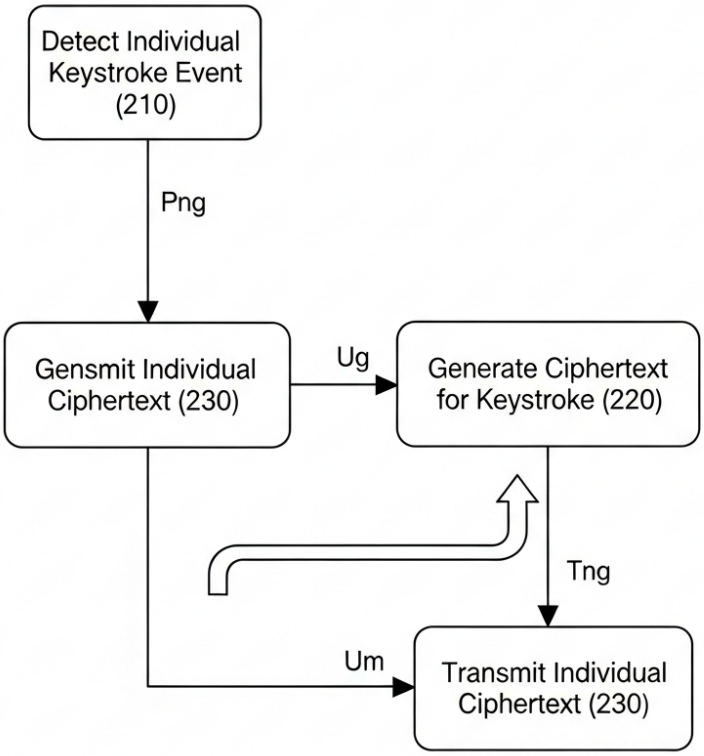


Figure 2: Flowchart depicting the method for processing and encrypting individual keystroke events.

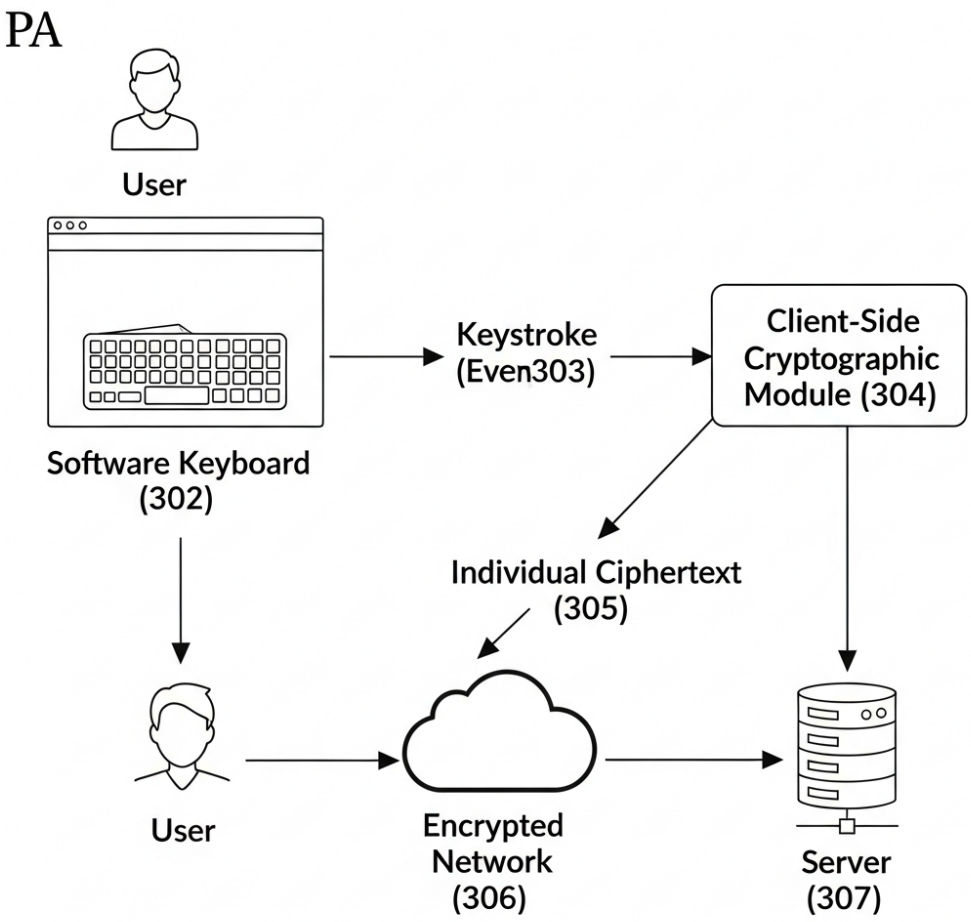


Figure 3: Conceptual diagram illustrating the application of individual keystroke encryption in a software keyboard environment.